

# Himanshu Nakrani

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## Professional Summary

Senior AI Engineer with 3+ years shipping production Generative AI for enterprise finance: agentic AI platforms, RAG pipelines, and Text-to-SQL systems serving 100+ users. Fine-tuned LLMs (GPT-4o mini, Llama 3.1) with end-to-end LLMops, pretrained a 1.12B-parameter model from scratch, and published two papers on Text-to-SQL.

## Technical Skills

**Generative AI & LLMs:** Large Language Models, RAG (Retrieval-Augmented Generation), Agentic AI, Multi-agent Systems, MCP (Model Context Protocol), AI Guardrails, Responsible AI, Text-to-SQL, Fine-tuning (LoRA/PEFT), Prompt Engineering, Model Optimization, vLLM (Inference Serving), LangChain, LangGraph, Hugging Face, OpenAI API, Azure OpenAI, Google ADK

**Backend Development:** FastAPI, REST APIs, SQLAlchemy, Microservices, API Design

**Databases:** SQL, Vector Databases (pgvector), Semantic Search, Query Optimization

**Machine Learning:** Deep Learning, NLP, Transformers, PyTorch, Data Pipelines, Model Deployment, Data Augmentation

**Cloud Platforms:** Azure, AWS, Google Cloud TPU, Oracle Cloud Infrastructure (OCI), Databricks

**Programming Languages:** Python, Java, SQL, Shell Scripting

**Developer Tools:** Git, Docker, Kubernetes, Harness, CI/CD, MLOps, LLMops, Model Observability, Testing, Evaluation Frameworks

## Professional Experience

### State Street Corporation

Hyderabad, India

Senior AI Engineer (Emerging Lead)

July 2023 – Present

- Spearheaded development of "**Agent Forge**", an enterprise **no-code agentic workflow platform** managing the complete agent lifecycle — visual design, deployment, monitoring, and evaluation of **multi-agent systems** — with centralized agent registration, discovery, and **RBAC (role-based access control)** for governance; adopted by **5+ business teams** running **50+ agent workflows** in production
- Designed and implemented OTEL-compliant **model observability**, advanced **AI guardrails** with input/output validation and content filtering, and automated evaluation pipelines for **Responsible AI** and regulatory compliance across all registered agents
- Architected and deployed "**Alpha Copilot**", an enterprise LLM-powered Text-to-SQL chatbot enabling natural language queries over structured financial data for 100+ internal users, with integrated LLM-generated data visualizations, interactive charts, and summaries accelerating decision-making
- Engineered scalable AI inference infrastructure using **FastAPI and SQLAlchemy**, implementing advanced caching and **latency optimization** techniques that reduced average response time by **75% (from 25-30s to 6-8s)**
- Developed production-ready **RAG system** for extracting actionable insights from fund prospectuses, engineering an optimized document pipeline with intelligent chunking, semantic embeddings, and vector database integration achieving sub-second retrieval times
- Led backend development for **WealthAI application**, building real-time portfolio narrative generation system that processes optimizer outputs and delivers personalized investment insights at scale with cross-functional Agile teams
- Fine-tuned **GPT-4o mini**, **GPT-4.1 mini** (Azure OpenAI) and **Llama 3.1-8B** (Databricks) using supervised fine-tuning and PEFT/LoRA, managing end-to-end LLMops pipeline including data augmentation, training, and inference optimization
- Researched advanced Text-to-SQL generation — schema-linking and few-shot prompting strategies — improving SQL accuracy by 25%
- Achieved **95%+ unit test coverage** and streamlined deployment workflows by architecting automated **CI/CD pipelines** with **Harness** and **Docker**, reducing production incidents

### State Street Corporation

Hyderabad, India

Software Development Intern

January 2023 – June 2023

- Led migration of legacy ideation platform from .NET Core to modern **FastAPI + ReactJS** stack, improving application performance by 40% and developer productivity
- Designed and implemented RESTful APIs with SQLite backend, ensuring seamless frontend integration and comprehensive test coverage

## Publications & Research

GoT4SQL-DA: A Graph-of-Thoughts Framework for Scalable Text-to-SQL Data Augmentation

FLLM 2025

GRAFT: Graph-of-Thoughts based Reasoning and Augmentation for Finetuning Text-to-SQL

CSCI 2025

## Key Projects

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**TinyMathReason-1B: Mathematical Reasoning LLM** | *JAX/MaxText, PyTorch, Cloud TPU, AMD MI300X, GCS, Python* | [GitHub](#)

- Pretrained a **1.12B parameter** Llama-style decoder-only transformer from scratch on a Google Cloud **TPU v4-32** cluster using **MaxText/JAX**, processing a **300B token** math-heavy dataset with a throughput of **66 TFLOP/s/device**
- Engineered a distributed data pipeline across parallelized cloud instances, processing **4 large-scale datasets** (FineWeb-Edu, OpenWebMath, Proof-Pile-2, Stack-Edu) with MinHash deduplication and sequence packing into optimized binary shards on GCS
- Built custom JAX/Orbax to Hugging Face checkpoint conversion utilities and implemented a two-stage post-training curriculum (**SFT, DPO/GRPO**) on an **AMD MI300X** GPU with custom tokenizer extensions for chain-of-thought reasoning

**Aegis: Agent Evaluation & Guardrails Harness** | *Google Gemini + ADK, FastAPI, PostgreSQL* | [GitHub](#)

- Built a production harness for agentic systems featuring evaluation frameworks, guardrails engine, and observability to ensure reliability of multi-agent workflows
- Leveraged Google ADK primitives (SequentialAgent, ParallelAgent, LoopAgent) to implement structured multi-agent patterns with shared state management
- Developed LLM-as-Judge evaluation and policy-based guardrails with real-time tracing, enabling safe and measurable deployment of agentic applications

## Education

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**Nirma University**  
*Bachelor of Technology in Electronics and Communication Engineering*

Ahmedabad, India  
*June 2019 – July 2023*

## Certifications

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**AWS Certified AI Practitioner** | [View Certificate](#)  
**Oracle Cloud Infrastructure Certified Generative AI Professional** | [View Certificate](#)